

Quantum Mechanics

Part-2

WAVE FUNCTION & SCHRÖDINGER EQUATION

Since $\Psi(x, t)$ represent the wave nature of matter, it should satisfy wave like equation
Particles in motion only shows wave nature, $\Psi(x, t)$ should satisfy travelling wave equation

$$\frac{\partial^2 y(x, t)}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y(x, t)}{\partial t^2}$$

Solution of the wave-equation :

Since the wave is travelling with velocity v , so the displacement $y(x, t)$ at x and $x \pm vt$ is same
Therefore, the amplitude at time t and $t \pm \frac{x}{v}$ will be the same

$$y(x, t) = f\left(t \pm \frac{x}{v}\right)$$

$$y(x, t) = A \cos \omega \left(t \pm \frac{x}{v}\right) \quad \text{or} \quad y(x, t) = A \sin \omega \left(t \pm \frac{x}{v}\right)$$

Most general form is $y(x, t) = Ae^{-i\omega\left(t \pm \frac{x}{v}\right)}$

WAVE FUNCTION & SCHRODINGER EQUATION

Associating this solution $y(x, t) = e^{-i\omega(t \pm \frac{x}{v})}$ with a free particle of mass m energy E and momentum p

$$E = h\nu = h \frac{\omega}{2\pi} \Rightarrow \omega = \frac{E}{\hbar} \qquad \frac{\omega}{\nu} = \frac{2\pi\nu}{\lambda\nu} = \frac{2\pi}{\lambda} = \frac{2\pi p}{h} = \frac{p}{\hbar}$$

Therefore we can write the free particle wave function as $\Psi(x, t) = e^{\frac{-i}{\hbar}(Et - px)}$

Note that

$$-i\hbar \frac{\partial \Psi(x, t)}{\partial x} = p \Psi(x, t) \quad \& \quad i\hbar \frac{\partial \Psi(x, t)}{\partial t} = E \Psi(x, t)$$

p & E are the eigen values of operators $-i\hbar \frac{\partial}{\partial x}$ & $i\hbar \frac{\partial}{\partial t}$ respectively

We can say that dynamical variables are represented by operators and $\hat{p} = -i\hbar \frac{\partial}{\partial x}$, $\hat{E} = i\hbar \frac{\partial}{\partial t}$

In the case of free particle $E = \frac{p^2}{2m}$

corresponding relation between operators $\hat{E}\Psi(x, t) = \frac{\hat{p}^2}{2m}\Psi(x, t)$

WAVE FUNCTION & SCHRODINGER EQUATION

$$\hat{E}\Psi(x, t) = \frac{\hat{p}^2}{2m}\Psi(x, t)$$

$$i\hbar \frac{\partial \Psi(x, t)}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi(x, t)}{\partial x^2}$$

The free particle wave function $\Psi(x, t) = e^{\frac{-i}{\hbar}(Et - px)}$ satisfy this equation

Concept can be generalized for particles influenced by a potential $U(x, t)$

The total energy relation for such system is $E = \frac{p^2}{2m} + U(x, t)$

Considering the corresponding operator relation we can write

$$i\hbar \frac{\partial \Psi(x, t)}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi(x, t)}{\partial x^2} + U(x, t)\Psi(x, t)$$

This is called the **Schrodinger Equation**

There is no proof that this equation will always determine the correct microscopic nature of a particle, however, violation of this relation for any system not yet observed !!!

SOLUTIONS OF SCHRÖDINGER EQUATION

If a particle is confined in special region due to some potential, the wave function should have following properties:

- The wave-function is continuous at the boundary.
- If the wave function comes out from boundary, derivative of the wave-function will also be continuous.
- If the boundary is very hard then wave-function is continuous but its derivative is not

SOLUTIONS OF SCHRODINGER EQUATION

A particle of mass m , under the influence of potential $U(x, t)$ is govern by

$$i\hbar \frac{\partial \Psi(x, t)}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi(x, t)}{\partial x^2} + U(x, t) \Psi(x, t) \quad \text{Or,} \quad i\hbar \frac{\partial \Psi(x, t)}{\partial t} = \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + U(x, t) \right] \Psi(x, t)$$

In 3D
$$i\hbar \frac{\partial \Psi(x, t)}{\partial t} = \left[-\frac{\hbar^2}{2m} \nabla^2 + U(x, t) \right] \Psi(x, t)$$

Separation of variable : If $U(x, t) = V(x)$, $\Psi(x, t) = \psi(x)\chi(t)$ 
$$i\hbar \frac{\partial \Psi(x, t)}{\partial t} = \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x) \right] \Psi(x, t)$$

$$i\hbar \frac{\partial \{\psi(x)\chi(t)\}}{\partial t} = \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} V(x) \right] \{\psi(x)\chi(t)\}$$

$$i\hbar \psi(x) \frac{\partial \chi(t)}{\partial t} = \chi(t) \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x) \right] \psi(x)$$



$$\boxed{\frac{i\hbar \frac{\partial \chi(t)}{\partial t}}{\chi(t)}}$$

Function of time only

$$\boxed{\frac{\left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x) \right] \psi(x)}{\psi(x)}}$$

Function of space only

$= E$

SOLUTIONS OF SCHRODINGER EQUATION

$$\frac{i\hbar \frac{\partial \chi(t)}{\partial t}}{\chi(t)} = E \quad \text{and} \quad \boxed{\frac{\left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x) \right] \psi(x)}{\psi(x)} = E} \Rightarrow \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x) \right] \psi(x) = E\psi(x)$$
$$\hat{H}\psi(x) = E\psi(x)$$

Time independent Schrodinger equation

Solving this equation we shall get **E** and **$\psi(x)$**

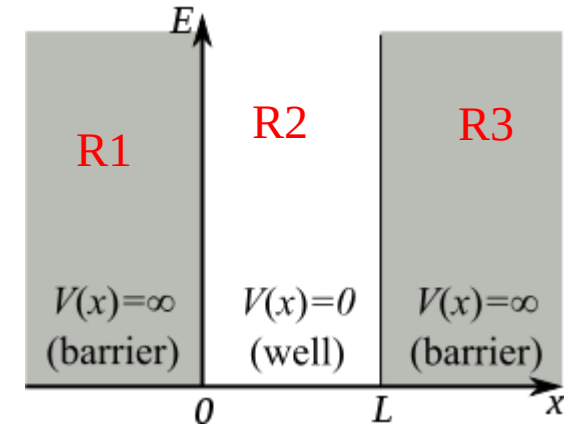
$$\frac{i\hbar \frac{\partial \chi(t)}{\partial t}}{\chi(t)} = E \Rightarrow i\hbar \frac{\partial \chi(t)}{\partial t} = E \chi(t) \Rightarrow \frac{\partial \chi(t)}{\chi(t)} = \frac{-i}{\hbar} E \Rightarrow \ln \chi(t) \approx \frac{-iEt}{\hbar} \Rightarrow \chi(t) \approx e^{-\frac{iEt}{\hbar}}$$

Total wave-function $\Psi(x, t) = \psi(x)\chi(t) = A \psi(x) e^{-\frac{iEt}{\hbar}}$

INFINITE POTENTIAL WELL

$$V(x) = \begin{cases} 0, & 0 \leq x \leq L \\ \infty & \text{else where} \end{cases}$$

The potential is time independent !!



There are three different regions R1, R2 and R3

In the regions R1, R3, wave function $\psi_1(x) = 0$ and $\psi_3(x) = 0$

In the regions R2, wave function $\psi_2(x)$ satisfies time independent Schrodinger equation

$$\left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x) \right] \psi_2(x) = E \psi_2(x) \quad \Rightarrow \quad \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + 0 \right] \psi_2(x) = E \psi_2(x)$$

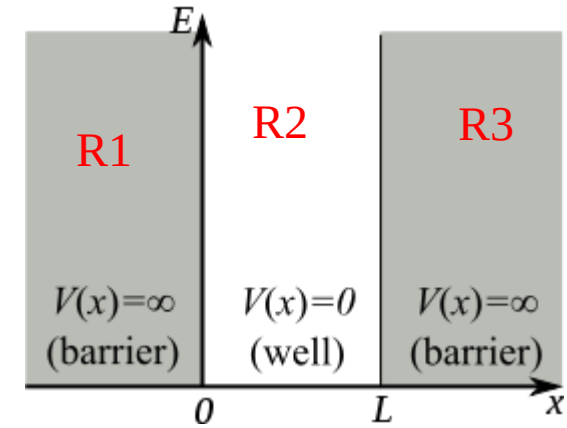
$$\Rightarrow \frac{\partial^2 \psi_2(x)}{\partial x^2} + \frac{2mE}{\hbar^2} \psi_2(x) = 0 \quad \Rightarrow \quad \frac{\partial^2 \psi_2(x)}{\partial x^2} + k^2 \psi_2(x) = 0 \quad \text{Where } k = \sqrt{\frac{2mE}{\hbar^2}}$$

INFINITE POTENTIAL WELL

$$V(x) = \begin{cases} 0, & 0 \leq x \leq L \\ \infty & \text{else where} \end{cases}$$

$$\frac{\partial^2 \psi_2(x)}{\partial x^2} + k^2 \psi_2(x) = 0 \quad k = \sqrt{\frac{2mE}{\hbar^2}}$$

$$\psi_2(x) = A \cos(kx) + B \sin(kx)$$



A and **B** are arbitrary constant will be determined from the boundary conditions

$\psi(x)$ is continuous at the boundary

$$\psi_2(0) = \psi_1(0) = 0 \quad \Rightarrow \quad A = 0 \quad \Rightarrow \quad \psi_2(x) = B \sin(kx)$$

$$\psi_2(L) = \psi_3(L) = 0 \quad \Rightarrow \quad \sin(kL) = 0 \quad \Rightarrow \quad kL = n\pi, \quad n = 1, 2, 3, \dots$$

$$k = \sqrt{\frac{2mE}{\hbar^2}} = \frac{n\pi}{L} \quad \Rightarrow \quad E = \frac{n^2 \pi^2 \hbar^2}{2mL^2}, \quad n = 1, 2, 3, \dots$$

INFINITE POTENTIAL WELL

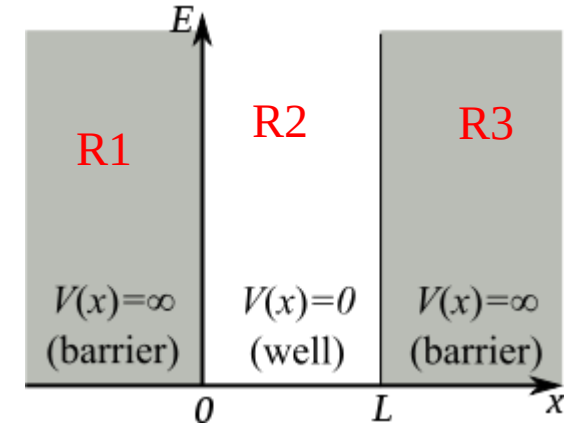
$$V(x) = \begin{cases} 0, & 0 \leq x \leq L \\ \infty & \text{else where} \end{cases}$$

$$\psi_2(x) = B \sin(kx) = \textcircled{B} \sin\left(\frac{n\pi}{L}x\right) \quad E = \frac{n^2\pi^2\hbar^2}{2mL^2}, \quad n = 1, 2, 3, \dots$$

$$\psi(x) = \begin{cases} \psi_2(x), & 0 \leq x \leq L \\ 0 & \text{else where} \end{cases}$$

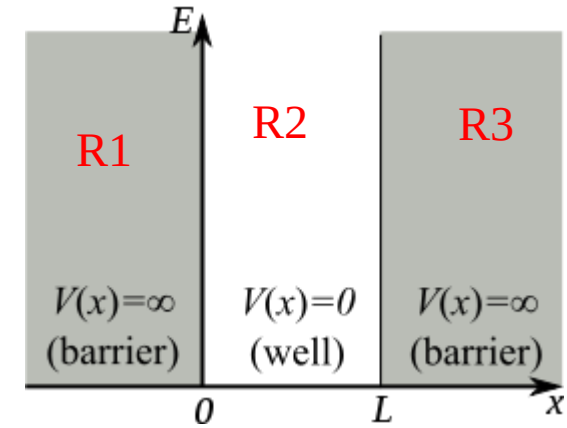
$$\int_{-\infty}^{\infty} \psi(x)^* \psi(x) dx = 1 \quad \Rightarrow \quad B^2 \int_0^L \sin^2\left(\frac{n\pi}{L}x\right) dx = 1 \quad \Rightarrow \quad B = \sqrt{\frac{2}{L}}$$

$$\psi(x) = \begin{cases} \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi}{L}x\right), & 0 \leq x \leq L \\ 0 & \text{else where} \end{cases} \quad E = \frac{n^2\pi^2\hbar^2}{2mL^2}, \quad n = 1, 2, 3, \dots$$



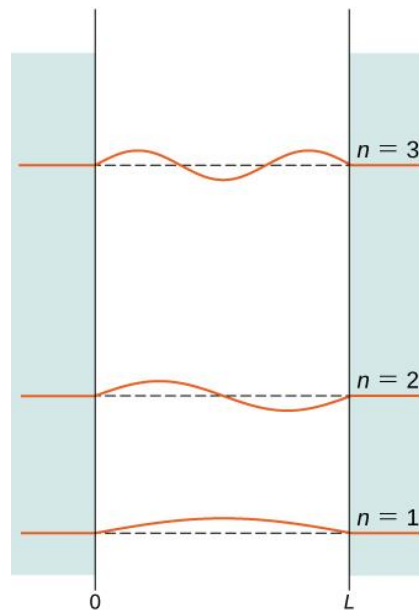
INFINITE POTENTIAL WELL

$$V(x) = \begin{cases} 0, & 0 \leq x \leq L \\ \infty & \text{else where} \end{cases}$$

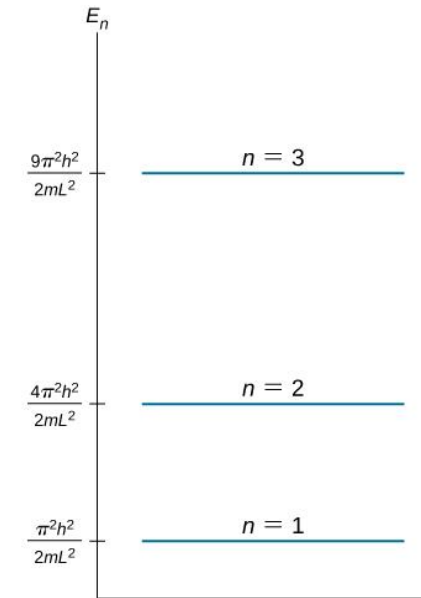


$$\psi(x) = \begin{cases} \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi}{L}x\right), & 0 \leq x \leq L \\ 0 & \text{else where} \end{cases}$$

$$E = \frac{n^2 \pi^2 \hbar^2}{2mL^2}, \quad n = 1, 2, 3, \dots$$



(a)



(b)

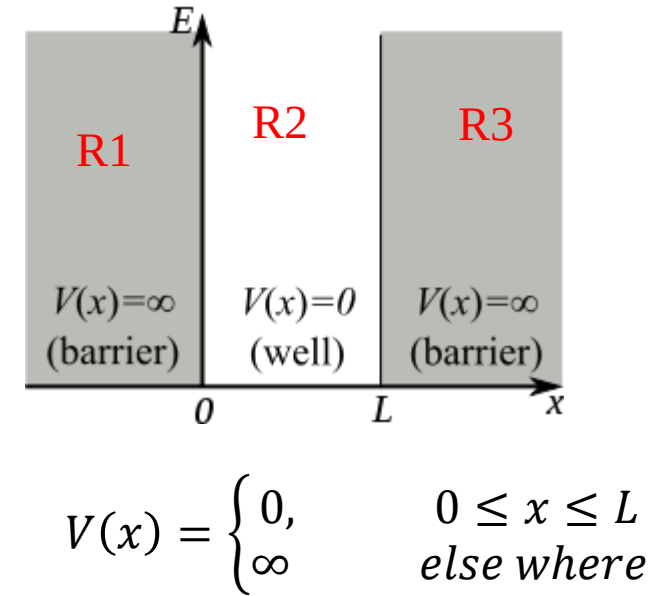
INFINITE POTENTIAL WELL

Q. What is the probability of finding the particle within the region $x = 0.45L$ & $x = 0.55L$?

$$\begin{aligned}\text{Ans.: } P &= \int_{x_1}^{x_2} \psi^* \psi dx = \frac{2}{L} \int_{x_1}^{x_2} \sin^2 \left(\frac{n\pi x}{L} \right) dx \\ &= \frac{x_2 - x_1}{L} - \frac{1}{2\pi n} \left(\sin \left(\frac{2n\pi x_2}{L} \right) - \sin \left(\frac{2n\pi x_1}{L} \right) \right)\end{aligned}$$

For $n = 1$, $P = 19.8 \%$

For $n = 2$, $P = 0.65 \%$

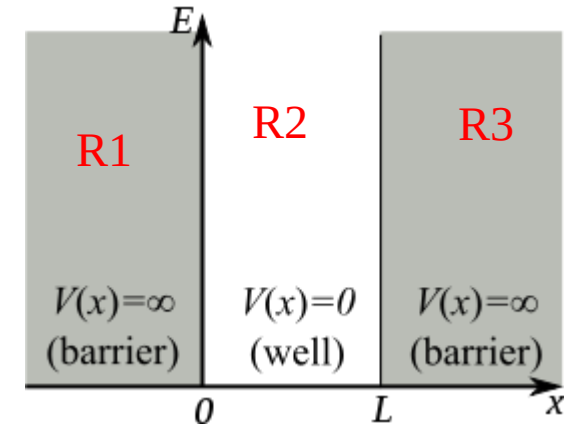


INFINITE POTENTIAL WELL

Q. What is the most probable position of the particle ?

Ans.:

$$\langle x \rangle = \int_{-\infty}^{\infty} \psi^* x \psi dx = \int_0^L x \sin^2 \left(\frac{n\pi x}{L} \right) dx = L/2$$

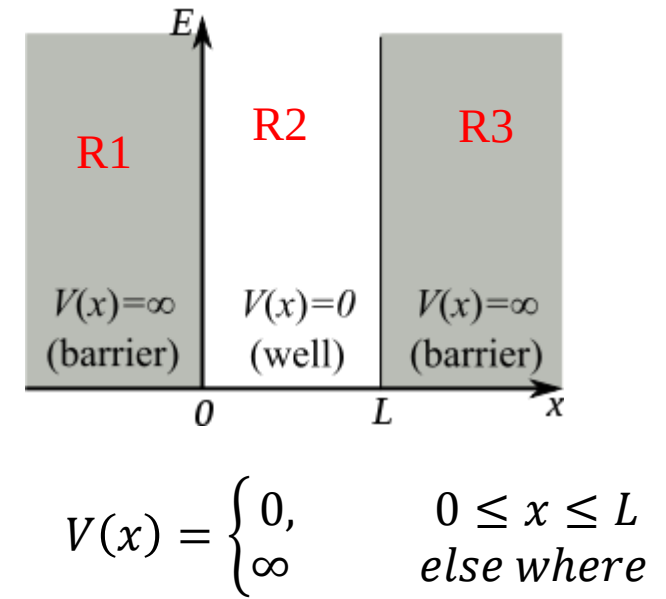


$$V(x) = \begin{cases} 0, & 0 \leq x \leq L \\ \infty & \text{else where} \end{cases}$$

INFINITE POTENTIAL WELL

Q. What is the expected value of particles momentum ?

$$\begin{aligned}\langle p \rangle &= \int_{-\infty}^{\infty} \psi^* \hat{p} \psi dx = \frac{2}{L} \int_0^L \sin\left(\frac{n\pi x}{L}\right) \left(-i\hbar \frac{\partial}{\partial x}\right) \sin\left(\frac{n\pi x}{L}\right) dx \\ &= 0\end{aligned}$$



INFINITE POTENTIAL WELL

Q. What is the Eigen function of momentum operator ?

$$\psi(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$$

$$\hat{p}\psi(x) = \sqrt{\frac{2}{L}} \left(-i\hbar \frac{\partial}{\partial x}\right) \sin\left(\frac{n\pi x}{L}\right) = -i\hbar \sqrt{\frac{2}{L}} \cdot \frac{n\pi}{L} \cos\left(\frac{n\pi x}{L}\right) \neq p\psi(x)$$

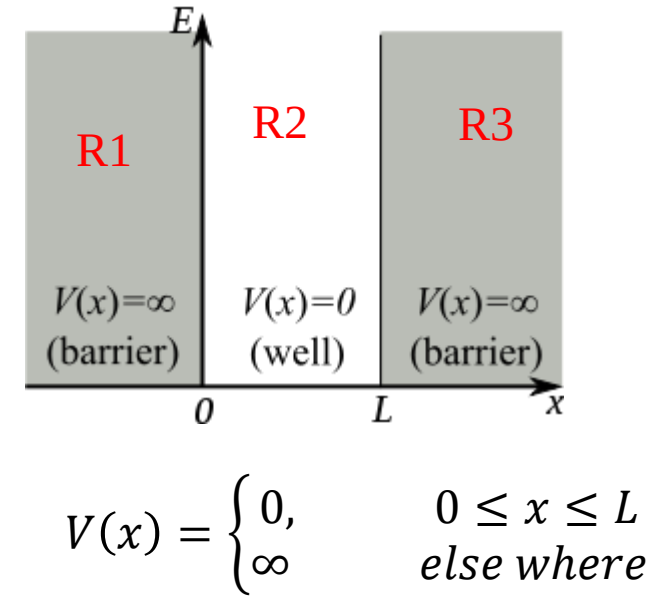
$$\psi(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) = \frac{\sqrt{\frac{2}{L}} (e^{i\frac{n\pi x}{L}} - e^{-i\frac{n\pi x}{L}})}{2i} = -\psi^+ + \psi^-$$

$$\psi^+ = \frac{i}{\sqrt{2L}} e^{i\frac{n\pi x}{L}}$$

$$\hat{p}\psi^+ = \left(-i\hbar \frac{\partial}{\partial x}\right) \left(\frac{i}{\sqrt{2L}} e^{i\frac{n\pi x}{L}}\right) = i \frac{n\pi}{L} \cdot (-i\hbar) \left(\frac{i}{\sqrt{2L}} e^{i\frac{n\pi x}{L}}\right) = \frac{n\pi\hbar}{L} \cdot \psi^+$$

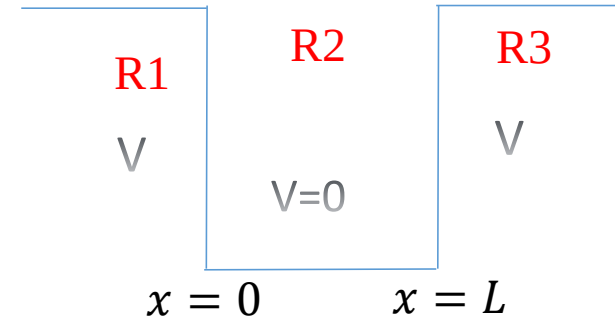
$$\psi^- = \frac{i}{\sqrt{2L}} e^{-i\frac{n\pi x}{L}}$$

$$\hat{p}\psi^- = \left(-i\hbar \frac{\partial}{\partial x}\right) \left(\frac{i}{\sqrt{2L}} e^{-i\frac{n\pi x}{L}}\right) = -i \frac{n\pi}{L} \cdot (-i\hbar) \left(\frac{i}{\sqrt{2L}} e^{-i\frac{n\pi x}{L}}\right) = -\frac{n\pi\hbar}{L} \cdot \psi^-$$



FINITE POTENTIAL WELL

$$V(x) = \begin{cases} 0, & 0 \leq x \leq L \\ V & \text{else where} \end{cases}$$



In the regions R1

$$\left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V \right] \psi_1(x) = E \psi_1(x)$$

➡ $\frac{\partial^2 \psi_1(x)}{\partial x^2} - a^2 \psi_1(x) = 0$ Where $a = \sqrt{\frac{2m(V-E)}{\hbar^2}}$

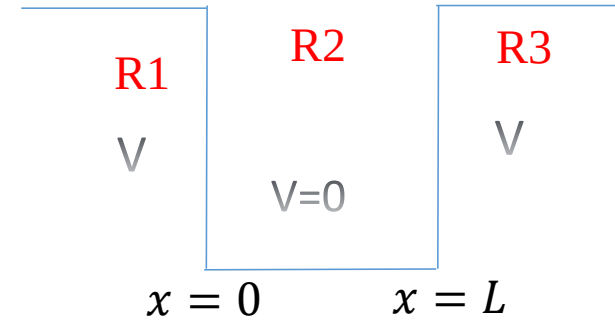
$\psi_1(x) = A e^{ax} + B e^{-ax}$ In this regions $\psi_1(x)$ should be finite as $x \rightarrow -\infty$ ➡ $B = 0$ ➡ $\psi_1(x) = A e^{ax}$

In the regions R3 $\left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V \right] \psi_3(x) = E \psi_3(x)$ ➡ $\frac{\partial^2 \psi_3(x)}{\partial x^2} - a^2 \psi_3(x) = 0$ Where $a = \sqrt{\frac{2m(V-E)}{\hbar^2}}$

$\psi_3(x) = C e^{ax} + D e^{-ax}$ In this regions $\psi_3(x)$ should be finite as $x \rightarrow \infty$ ➡ $C = 0$ ➡ $\psi_3(x) = D e^{-ax}$

FINITE POTENTIAL WELL

$$V(x) = \begin{cases} 0, & 0 \leq x \leq L \\ V & \text{else where} \end{cases}$$



In the regions R2

$$\left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + 0 \right] \psi_3(x) = E \psi_3(x)$$

➔ $\frac{\partial^2 \psi_3(x)}{\partial x^2} + k^2 \psi_3(x) = 0$ Where $k = \sqrt{\frac{2mE}{\hbar^2}}$

$$\psi_2(x) = E \cos(kx) + F \sin(kx)$$

$$\psi_1(x) = A e^{ax}$$

$$\psi_2(x) = E \cos(kx) + F \sin(kx)$$

$$\psi_3(x) = D e^{-ax}$$

SOLUTIONS OF SCHRÖDINGER EQUATION

If a particle is confined in special region due to some potential, the wave function should have following properties:

- The wave-function is continuous at the boundary.
- If the wave function comes out from boundary, derivative of the wave-function will also be continuous.
- If the boundary is very hard then wave-function is continuous but its derivative is not

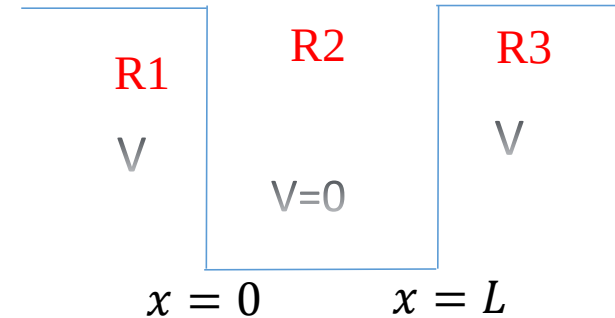
FINITE POTENTIAL WELL

$$\psi_1(x) = A e^{ax}$$

$$\psi_2(x) = B \cos(kx) + C \sin(kx)$$

$$\psi_3(x) = D e^{-ax}$$

Where $a = \sqrt{\frac{2m(V-E)}{\hbar^2}}$
 $k = \sqrt{\frac{2mE}{\hbar^2}}$



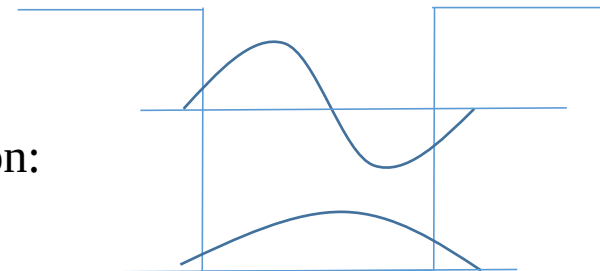
We need to find the value of arbitrary constant **A**, **B**, **C** and **D** and also **k** or **a** from the boundary conditions

Wave function continuous at the boundary: $\psi_1(0) = \psi_2(0)$ and $\psi_2(L) = \psi_3(L)$

Derivative of wave function continuous at the boundary: $\psi_1'(0) = \psi_2'(0)$ and $\psi_2'(L) = \psi_3'(L)$

$$\tan kL = \left(\frac{2ka}{a^2 - k^2} \right)$$

Expected wave function:



POTENTIAL BARRIER

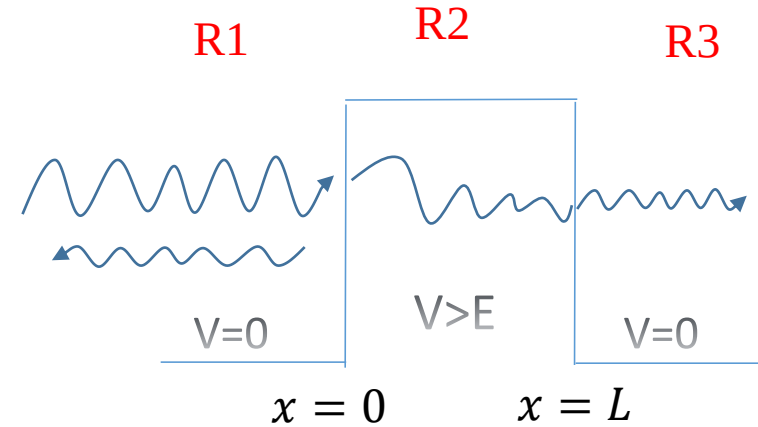
In the regions R1: $\left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + 0 \right] \psi_1(x) = E \psi_1(x)$

$$\psi_1(x) = A e^{ikx} + B e^{-ikx} \quad k = \sqrt{\frac{2mE}{\hbar^2}}$$

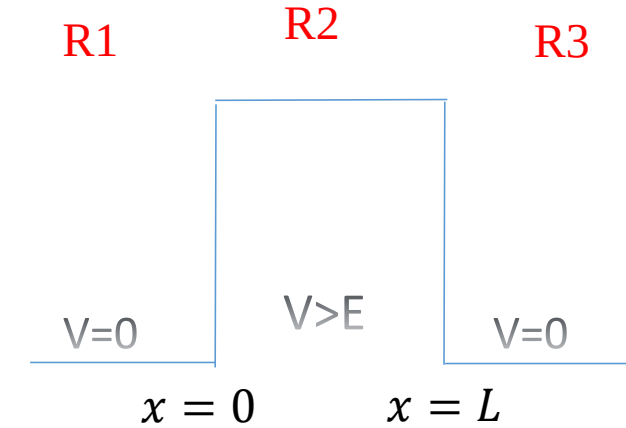
In the regions R3: $\psi_3(x) = E e^{ikx}$

In the regions R2: $\left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V \right] \psi_2(x) = E \psi_2(x)$

$$\psi_2(x) = C e^{ax} + D e^{-ax} \quad \text{Where } a = \sqrt{\frac{2m(V-E)}{\hbar^2}}$$



POTENTIAL BARRIER



$$\psi_1(x) = A e^{ikx} + B e^{-ikx}$$

$$\psi_2(x) = C e^{ax} + D e^{-ax}$$

$$\psi_3(x) = E e^{ikx} \quad k = \sqrt{\frac{2mE}{\hbar^2}} \quad \& \quad a = \sqrt{\frac{2m(V-E)}{\hbar^2}}$$

$$\psi_1(0) = \psi_2(0) \Rightarrow A + B = C + D$$

$$\psi_1'(0) = \psi_2'(0) \Rightarrow ik(A - B) = a(C - D)$$

$$\psi_2(L) = \psi_3(L) \Rightarrow C e^{aL} + D e^{-aL} = E e^{ikL}$$

$$\psi_2'(L) = \psi_3'(L) \Rightarrow a(C e^{aL} - D e^{-aL}) = ikE e^{ikL}$$

➤ We need to find the value of arbitrary constant **A**, **B**, **C**, **D** and **E** from four equation !!!

➤ Instead we calculate **B/A**, **C/A**, **D/A** and **E/A**

➤ The transmission probability $T = \frac{E^* E}{A^* A}$

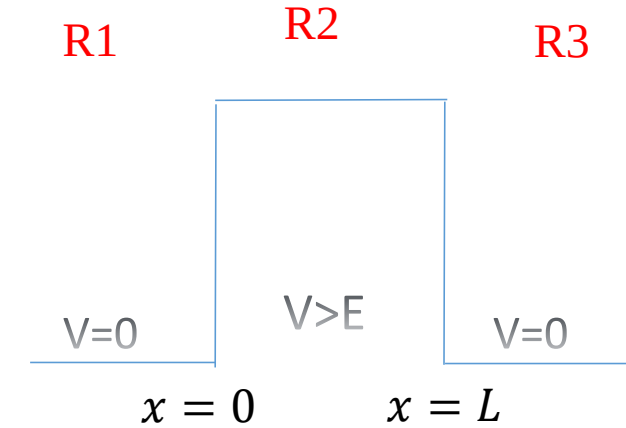
$$\frac{A}{E} = \frac{1}{4} \left(1 - \frac{ia}{k} \right) \left(1 + \frac{ik}{a} \right) e^{(ik-a)L} + \frac{1}{4} \left(1 + \frac{ia}{k} \right) \left(1 - \frac{ik}{a} \right) e^{(ik+a)L}$$

For a large V , $a = \sqrt{\frac{2m(V-E)}{\hbar^2}} \gg 1$, and $e^{-aL} \rightarrow 0$, we can neglect the first term

POTENTIAL BARRIER

$$\frac{A}{E} = \frac{1}{4} \left(1 + \frac{ia}{k} \right) \left(1 - \frac{ik}{a} \right) e^{(ik+a)L}$$

$$T = \frac{E^*E}{A^*A} = \frac{1}{\left(\frac{1}{4} + \frac{1}{16} \frac{a^2}{k^2} \right)} e^{-2aL} \approx e^{-2aL}$$



- We need to find the value of arbitrary constant **A**, **B**, **C**, **D** and **E** from four equation !!!
- Instead we calculate **B/A**, **C/A**, **D/A** and **E/A**
- The transmission probability $T = \frac{E^*E}{A^*A}$

TUNNEL EFFECT

All the animations and explanations on
www.toutestquantique.fr

Problems

Q.1 The lowest energy of a particle in an infinite one-dimensional well is 4.4 eV. If the width of the well is doubled, what is its lowest energy ?

Ans.:

$$E_1 = \frac{h^2}{8mL^2} = 4.4 \text{ eV}$$

With $L' = 2L$,

$$\begin{aligned} E'_1 &= \frac{h^2}{8mL'^2} = \frac{h^2}{8m(2L)^2} = \frac{1}{4} \frac{h^2}{8mL^2} \\ &= \frac{1}{4} (4.4 \text{ eV}) = 1.1 \text{ eV} \end{aligned}$$

Problems

Q.2 An electron is trapped in an infinite well of width 0.120 nm. What are the three longest wavelengths permitted for the electron's de Broglie waves?

Ans.:

Possible wavelengths should satisfy the condition $L = \frac{n\lambda_n}{2}$

$$\lambda_n = \frac{2L}{n}, \quad n = 1, 2, 3 \dots \dots$$

$$\lambda_1 = \frac{2(0.120 \text{ nm})}{1} = 0.240 \text{ nm}$$

$$\lambda_2 = \frac{0.240 \text{ nm}}{2} = 0.120 \text{ nm}$$

$$\lambda_3 = \frac{0.240 \text{ nm}}{3} = 0.080 \text{ nm}$$

Problems

Q.3 What is the minimum energy of a neutron ($mc^2 = 940 \text{ MeV}$) confined to a region of space of nuclear dimensions ($1.0 \times 10^{-14} \text{ m}$)?

Ans.:

With $L = 1.0 \times 10^{-14} \text{ m} = 10 \text{ fm}$,

$$E_1 = \frac{h^2}{8mL^2} = \frac{(hc)^2}{8(mc^2)L^2} = \frac{(1240 \text{ MeV} \cdot \text{fm})^2}{8(940 \text{ MeV})(10 \text{ fm})^2} = 2.0 \text{ MeV}$$

Problems

Q.4 A particle is described by the wave function $\psi(x) = b(a^2 - x^2)$ for $-a \leq x \leq +a$ and $\psi(x) = 0$ for $x \leq -a$ and $x \geq +a$, where a and b are positive real constants.

- (a) Using the normalization condition, find b in terms of a .
- (b) What is the probability to find the particle at $x = +a/2$ in a small interval of width $0.010a$?
- (c) What is the probability for the particle to be found between $x = +a/2$ and $x = +a$?

Ans.:

(a) The regions with $x < a$ and $x > a$ do not contribute to the normalization. The normalization integral is

$$\int |\psi(x)|^2 dx = 1 \quad \Rightarrow \quad \int_{-a}^{+a} b^2 (a^2 - x^2)^2 dx = 1 \quad \Rightarrow \quad b^2 \left(2a^5 - \frac{4a^5}{3} + \frac{2a^5}{5} \right) = 1 \quad \text{or} \quad b = \sqrt{\frac{15}{16a^5}}$$

$$(b) \quad P(x) dx = |\psi(x)|^2 dx = b^2 (a^2 - x^2)^2 dx$$

$$\text{with } x = +a/2 \text{ and } dx = 0.010a \quad P(x) dx = \frac{15}{16a^5} \left(\frac{a^2}{4} - a^2 \right)^2 (0.010a) = 0.0053$$

$$(c) \quad P(a/2 : a) = \int_{a/2}^a |\psi(x)|^2 dx = \int_{a/2}^a b^2 (a^2 - x^2)^2 dx = \frac{15}{16a^5} \left(a^4 x - 2a^2 \frac{x^3}{3} + \frac{x^5}{5} \right) \Bigg|_{a/2}^a = 0.104$$

Problems

Q.5 . A particle in an infinite well is in the ground state with an energy of 1.26 eV. How much energy must be added to the particle to reach the second excited state ($n = 3$)? The third excited state ($n = 4$)?

Ans.:

$$E = \frac{n^2 \pi^2 \hbar^2}{2mL^2}, \quad n = 1, 2, 3, \dots \dots \dots$$

$$E_n = n^2 E_1 \quad \& \quad E_1 = 1.26 \text{ eV}$$

$$\Delta E_3 = E_3 - E_1 = 9E_1 - E_1 = 8E_1 = 8(1.26 \text{ eV}) = 10.1 \text{ eV}$$

$$\Delta E_4 = E_4 - E_1 = 16E_1 - E_1 = 15E_1 = 15(1.26 \text{ eV}) = 18.9 \text{ eV}$$

WAVE FUNCTION & SCHRODINGER EQUATION

- ❖ Wave properties of matter is represented by the wave function $\Psi(x, t)$
- ❖ In general $\Psi(x, t)$ is complex function
- ❖ $\Psi^*(x, t)\Psi(x, t)$ is a positive definite quantity associated with the probability of finding the particle at the position x at a time t .
- ❖ $\int_{-\infty}^{\infty} \Psi^*(x, t)\Psi(x, t)dx \geq 0 \Rightarrow$ is the total probability (finding the particle anywhere in the space)
- ❖ $\int_{-\infty}^{\infty} \Psi^*(x, t)\Psi(x, t)dx = 0 \Rightarrow$ No particle exists.
- ❖ $\int_{-\infty}^{\infty} \Psi^*(x, t)\Psi(x, t)dx = 1 \Rightarrow$ Normalized wave function.

WAVE FUNCTION & SCHRODINGER EQUATION

Well behaved $\Psi(x, t)$:

- ❖ $\Psi(x, t)$ must be single valued and continuous everywhere
- ❖ $\frac{\partial \Psi(x, t)}{\partial x}$ must be continuous and single valued
- ❖ $\Psi(x, t)$ must be normalizable $\Rightarrow \int_{-\infty}^{\infty} \Psi^*(x, t) \Psi(x, t) dx$ is finite
i.e. $\Psi(x, t) \rightarrow 0$ as $x \rightarrow \pm\infty$

WAVE FUNCTION & SCHRÖDINGER EQUATION

Let us associate $\Psi(x, t) = Ae^{-i\omega(t \pm \frac{x}{v})}$ with a free particle of energy E and momentum p

$$\omega = 2\pi\nu = \frac{2\pi E}{h} = \frac{E}{\hbar} \quad \text{and} \quad \frac{\omega}{v} = \frac{2\pi\nu}{\lambda\nu} = \frac{2\pi}{\lambda} = \frac{2\pi p}{h} = \frac{p}{\hbar}$$

$$\Psi(x, t) = Ae^{-\frac{i}{\hbar}(Et \pm px)}$$